# U.S. Antimony Supply Resilience

Strategic options for a hypothetical defense manufacturer, 2027–2031

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Illustrative procurement model · 8-slide edition

# Qualify a second supply lane, then hold five months of stock

Conditional balanced policy for 100 t/year of contained antimony-equivalent demand

**45%**

Target share from an independently qualified alternative international supply lane

**5 months**

Provisional buyer-qualified reserve: $2.30m at the historical USGS metal proxy

**30%**

Future domestic offtake target; availability begins only with verified qualified output

### 2030 severe interruption

## 0 t unfilled under modeled supplier assumptions

2027–29: the modeled minimum buffer is 4.91 months for ≤5 t unfilled. Five months still leaves 4.35 t unfilled before domestic qualification.

USGS historical price proxy; model inputs and assumptions are documented in the public evidence register.

Evidence and assumptions

SUPPLY BASELINE

# National import reliance does not establish a buyer's supplier mix

USGS estimates and historical U.S. antimony import origin, different denominators

**91%**

U.S. net import reliance, 2025 estimate

**55%**

China share of U.S. metal + oxide import origins, 2021–2024

BUYER-SPECIFIC DILIGENCE

| | |
|---|---|
| **Form + grade** | Contained-Sb conversion and specification |
| **Traceability** | Mine, smelter and refinery location |
| **Qualification** | Test results, supply independence and acceptance |

China's item-specific export licensing is a separate sourcing gate. EXIM's $2.906bn Stibnite authorization and an INL pilot still require buyer-level qualification evidence.

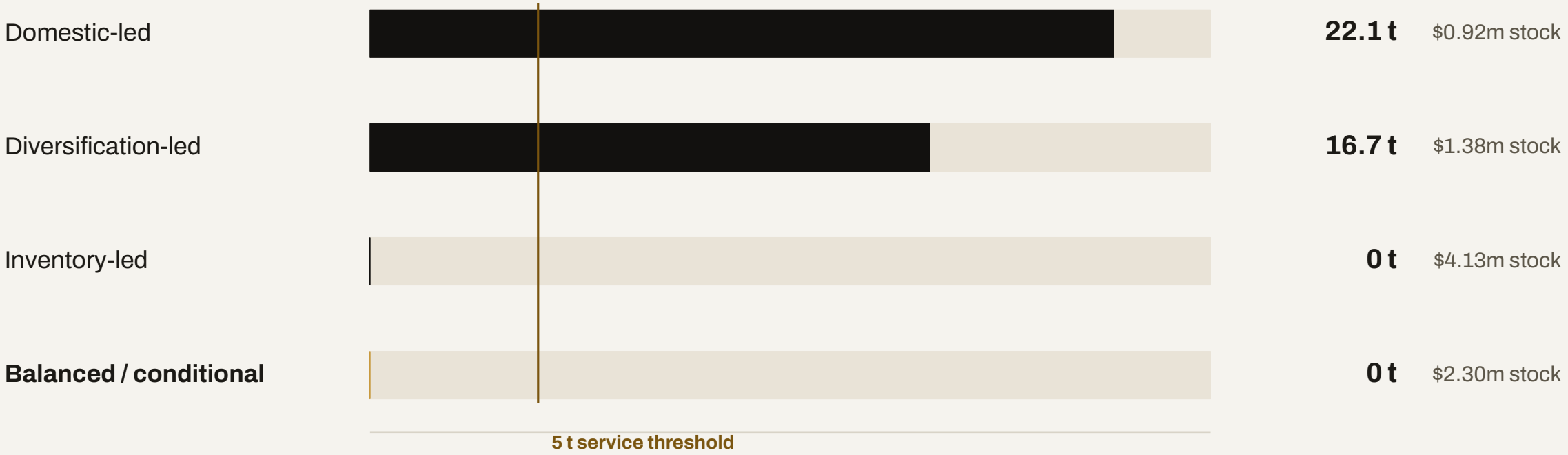
USGS MCS 2026 | EXIM transaction register | INL pilot | MOFCOM No. 33

FOUR-STRATEGY COMPARISON

# Only the balanced strategy meets both modeled 2030 constraints

Severe eight-month disruption; 100 t/year buyer; cash cap $3m; service tolerance ≤5 t unfilled

UNFILLED ANTIMONY-EQUIVALENT (T)




Inventory-led passes continuity but exceeds the illustrative $3m stock limit; balanced passes both only under specified qualification assumptions.

Model register / 2030 comparison

# Five months frees $1.84m of stock cash versus nine months

2030 severe case; annual incremental expense includes sourcing premiums, carry and spot markup

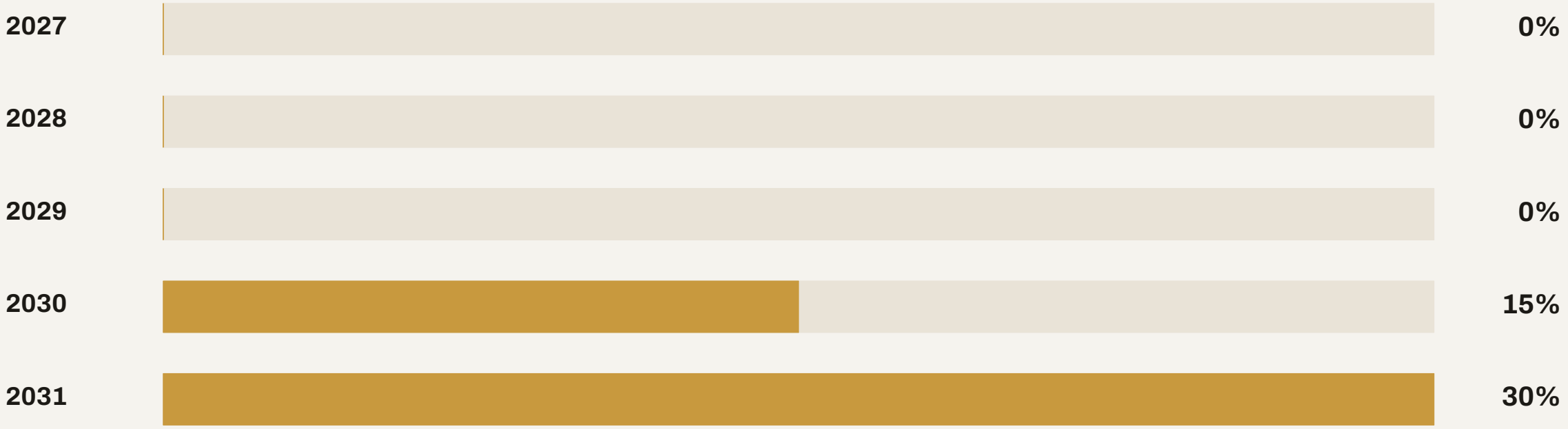
| | BALANCED / CONDITIONAL | INVENTORY-LED |
|---|---|---|
| **Stock months** | 5 months | 9 months |
| **Inventory capital** | $2.296m | $4.134m |
| **Modeled annual increment** | $813k | $703k |
| **2030 residual unfilled** | 0 t | 0 t |

**Trade-off: lower initial working capital at approximately $110k more annual modeled expense**

Initial inventory cash is an asset, not an annual incremental expense; all amounts use hypothetical buyer and historical price inputs.

# The domestic target becomes available only after qualification

Modeled ramp to 30% target share; all values are illustrative buyer sourcing shares




Until qualified domestic deliveries are available, that target volume is assigned to the incumbent sourcing lane in the model.

Actual schedules, plant status, processing specifications and enforceable offtake terms would control a real procurement ramp.

# The five-month policy is sensitive to supplier independence

Residual unfilled contained-Sb equivalent in each distinct adverse scenario

| SEVERE TEST | MODELED GAP | OUTCOME AGAINST ≤5 t |
|---|---|---|
| **2027: no qualified alternative** | 16.50 t | Above tolerance |
| **2030: alternative lane 75% curtailed** | 5.70 t | Above tolerance |
| **2030: curtailment + domestic delay** | 12.45 t | Above tolerance |
| **2030: 12-month interruption** | 15.15 t | Above tolerance |
| **Action: strengthen alternate refinery qualification and retain a sixth-month procurement option** | | |

Each test is a separate modeled shock, not an expected forecast or cumulative five-year loss.

EXECUTION GATES

# Procurement, Finance and Legal each control a material assumption

Decision rights before firming purchases, inventory funding or domestic offtake

| OWNER | VERIFICATION REQUIRED | TRIGGER |
|---|---|---|
| **Procurement + Engineering** | Grade, chain of custody, independent refining, delivery tests | Qualify alternative 45% lane |
| **Finance** | Actual quoted price, carry, storage, working-capital ceiling | Approve five-month reserve |
| **Strategy + Legal** | Financing close, processing stage, product acceptance, contract remedies | Stage conditional domestic offtake |

USGS MCS 2026 · EXIM register · Perpetua 10-Q · INL pilot · MOFCOM No. 33

MOFCOM No. 46 · MOFCOM No. 72 · Model register · USAC 10-Q · DLA stockpile

ILLUSTRATIVE BUYER CASE · SOURCE CUTOFF 4 NOV 2026 · INDEPENDENT SMPT ANALYSIS

Read case / model